Umpire rotation and inherited integrity in an evolutionary model of institutional corruption

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Abstract

Institutions that punish free riders can sustain cooperation, but the officers who run them may become corrupt. Lee, Iwasa, Dieckmann and Sigmund (2019) showed that social learning then produces recurring cycles of corruption. We reproduce their model and extend it with term-limited officers (umpires) who are replaced at a fixed rate by newcomers whose integrity may be inherited from the incumbents. We reproduce the paper's analytical benchmarks exactly and its simulations closely, and we document six inconsistencies in the paper and its supplementary information. In the large-population limit, rotation removes corruption cycles through a saddle-node bifurcation on the cycle at a critical rotation rate. We derive an approximation for this rate and show that it is proportional to the exploration rate. In stochastic simulations of small populations, rotation raises the time-averaged share of honest umpires from 0.669 to 0.978 and makes corruption episodes rarer. Rotation also lowers vigilance, but contrary to our hypothesis the remaining episodes are shallower, not deeper. Inheritance weakens rotation in all three variants studied; linear inheritance is equivalent to slower fresh rotation. Conformist inheritance produces bistability only when newcomers almost purely copy the majority, and delayed inheritance does not destabilize the honest state. Replacing whole cohorts at once is more effective than gradual replacement at the same average rate.

Keywords: corruption, evolutionary game theory, institutions, term limits, replicator dynamics, bifurcation

What is reproduced and what is new. Sections 2.1, 3 and 4.1 reproduce Lee et al. [1]. The rotation model (Section 2.2), its analysis (Section 4.2) and all results on hypotheses H1 to H6 (Section 4.3) are new. We checked the model against the supplementary information (SI) of Lee et al. The SI states the exploration and fixation rules, which our model follows, but not the order of individual updates; that part remains our assumption and is marked as such (Appendices B and E).

1 Introduction

Cooperation in groups is often protected by institutions that punish those who cheat. Courts, regulators, referees and auditors make defection costly, and in doing so they turn a social dilemma into a situation where cooperation pays. Ostrom described such institutions as tools that let people overcome collective-action problems [2]. Evolutionary game theory has shown how sanctioning institutions can arise and persist through social learning, for example when individuals choose whether to contribute to a punishment pool [3], and how institutional rewards and penalties can be combined efficiently [4, 5].

Institutions are run by people, and people can be bribed. Corruption of enforcers is a central topic in political economy [6] and has been studied experimentally [7]. Lee, Iwasa, Dieckmann

and Sigmund [1] built a minimal evolutionary model of this problem. Pairs of players play a donation game supervised by an umpire. An honest umpire fines defectors; a corrupt umpire takes a bribe from them instead. Players can pay a small cost to learn the type of their umpire and refuse to play under one that does not suit them. Both players and umpires learn by imitating more successful peers. The model produces cycles: when umpires are honest, checking them seems wasteful, so optimistic players spread; corrupt umpires then gain from bribes and take over; cooperation collapses until prudent players return, at which point honest umpires earn more and recover. In large populations with more exploration, the cycles are damped and corruption persists at an intermediate level [1]. Related models study corrupt enforcers in pool punishment [8], enforcer education against illegal logging [9], and supervision of umpires by players [10] or by external services [11]. Cycles of corruption have also been described in political models [12].

One common policy response to corruption is to limit how long officials stay in office. Staff rotation has been tested in laboratory experiments [13, 14], and term limits have been linked to corruption in empirical work [15, 16]. The effect is not obvious. New officials may not yet be part of corrupt networks, but they may also learn the norms of the body they join. Rotation is also everyday practice in many small institutions, such as student organizations whose leaders change every year. In such settings the integrity of the institution depends both on how often its members change and on what newcomers learn from their predecessors.

To our knowledge, rotation of enforcers has not been analyzed in the model of Lee et al. or in the other evolutionary-game models of corruption that we found (Appendix F). Here we first reproduce the model from the main text and its supplementary information (SI). We then add umpire rotation: each umpire is replaced at a fixed rate, and the newcomer is honest with a probability that is either fixed (a fresh cohort) or inherited from the current umpires. Inheritance can be linear, conformist, or delayed by one term. We state six hypotheses in advance (H1 to H6), fix the metrics and decision rules before running the experiments, and evaluate each hypothesis in the deterministic large-population limit (mean field) and in finite-population simulations.

Our contributions are as follows. We reproduce the analytical benchmarks of Lee et al. exactly, including an independent derivation of their rare-exploration Markov chain, and their simulations closely. We document six inconsistencies in the paper and its SI, none of which changes the paper's conclusions. We show that rotation removes the corruption cycle not through a Hopf bifurcation but through a saddle-node bifurcation on the limit cycle, and we derive an approximation for the critical rotation rate. We find that rotation raises honesty and lowers vigilance as expected, but that the remaining corruption episodes are not deeper, so a "vigilance paradox" does not appear. We find that inheritance weakens rotation, that conformist inheritance creates bistability only in an extreme regime, and that delayed inheritance does not destabilize the honest state. Finally, we find that replacing whole cohorts at once is more effective than gradual replacement.

2 Model

2.1 Base model (Lee et al. 2019)

This subsection restates the model of Lee et al. [1]; all credit for it belongs to them. Source locations for every element are listed in the repository file `SPEC.md`.

There are M players and N umpires. Two random players meet and are assigned a random umpire. Each player follows one of four strategies: optimistic cooperator (OC), prudent cooperator (PC), optimistic defector (OD) or prudent defector (PD). A prudent player pays h to learn whether the umpire is honest. A PD refuses to play under an honest umpire and a PC refuses to play under a corrupt one. If both players proceed, each pays a fee f to the umpire. A corrupt umpire also takes a bribe B from each defector. The players then play a donation game with benefit b and cost c, and an honest umpire fines each defector A. The resulting payoff matrices

for the strategy order (OC, PC, OD, PD) are

$$Q_H = \begin{pmatrix} b-c-f & b-c-f & -c-f & 0 \\ b-c-f-h & b-c-f-h & -c-f-h & -h \\ b-f-A & b-f-A & -f-A & 0 \\ -h & -h & -h & -h \end{pmatrix}, \tag{1}$$

$$Q_C = \begin{pmatrix} b-c-f & 0 & -c-f & -c-f \\ -h & -h & -h & -h \\ b-f-B & 0 & -f-B & -f-B \\ b-f-B-h & -h & -f-B-h & -f-B-h \end{pmatrix}. \tag{2}$$

An honest umpire earns $2f$ unless a PD walks away; a corrupt umpire earns $2f$ plus B per defector unless a PC walks away. With $x = (x_1, \ldots, x_4)$ the frequencies of OC, PC, OD, PD and y the fraction of honest umpires, the expected umpire incomes are $P_H = 2f(1-x_4)^2$ and $P_C = 2f(1-x_2)^2 + 2B(x_3+x_4)(1-x_2)$, so that

$$P_H - P_C = 2f(x_2 - x_4)(1 + x_1 + x_3) - 2B(x_3 + x_4)(1 - x_2), \tag{3}$$

as stated by Lee et al. Their parameter conditions are $f + h < b - c$, $c + h < A$, $B + h < c$ and $h < \sqrt{f(f+B)}$. The last one guarantees that honest umpires can invade a population of corrupt umpires and prudent cooperators. We use their base values $b = 1$, $c = 0.5$, $f = B = 0.2$, $h = 0.1$ and $A = 2$ (or $A = 0.7$).

In large populations the dynamics are the replicator-mutator equations

$$\dot{x}_i = x_i\big[(Q(y)x)_i - x^\top Q(y)x\big] - \mu x_i + \mu\frac{1-x_i}{3}, \qquad i = 1, \ldots, 4, \tag{4}$$

$$\dot{y} = y(1-y)\big[P_H(x) - P_C(x)\big] - \nu y + \nu(1-y), \tag{5}$$

with exploration rates μ (players) and ν (umpires) and $Q(y) = yQ_H + (1-y)Q_C$. The printed equation has the weights of Q_H and Q_C swapped; we use the reading that is consistent with y being the honest fraction (Appendix B).

In finite populations, individuals revise their strategies one at a time. With probability μ (players) or ν (umpires) the revising individual switches to another strategy at random; otherwise it compares its payoff P_X with that of a random role model Y from its own population and copies Y with probability $[1 + \exp(s(P_X - P_Y))]^{-1}$, where s is the selection strength.

2.2 Rotation and inherited integrity (new)

We add term limits for umpires. This is our design (an assumption, not part of Lee et al.). Each umpire is replaced at rate ρ by a newcomer who is honest with probability p_{new}. In the mean field this adds one term to Eq. (5):

$$\dot{y} = y(1-y)\big[P_H - P_C\big] - \nu y + \nu(1-y) + \rho\,(p_{\text{new}} - y). \tag{6}$$

We study four rules for the newcomer's honesty:

$$\begin{aligned}
&\text{V0 (fresh cohort):} && p_{\text{new}} = p_0, \\
&\text{V1 (linear inheritance):} && p_{\text{new}} = (1-\kappa)p_0 + \kappa y, \\
&\text{V2 (conformist inheritance):} && p_{\text{new}} = (1-\kappa)p_0 + \kappa\, g(y), \quad g(y) = \frac{y^k}{y^k + (1-y)^k},\ k = 2, \\
&\text{V3 (delayed inheritance):} && p_{\text{new}}(t) = (1-\kappa)p_0 + \kappa\, y(t - T_{\text{lag}}).
\end{aligned}$$

Here p_0 is the honesty of an untrained newcomer (default 1) and $\kappa \in [0, 1]$ is the weight of inheritance. In V3 the newcomer learns from the body as it was one term earlier; by default

$T_{\mathrm{lag}} = 1/\rho$, the mean term length. We also study pulsed rotation, in which a fraction τ of umpires is replaced every T time units, so that $y \to (1-\tau)y + \tau p_{\mathrm{new}}$ at each pulse and the average rate is $\rho = \tau/T$. In the finite-population model, each umpire is replaced at rate ρ per generation (continuous) or $\lfloor \tau N \rceil$ randomly chosen umpires are replaced every T generations (pulses).

Two remarks help interpretation. First, for V1 the rotation term equals $\rho(1-\kappa)(p_0 - y)$, so linear inheritance is identical to fresh rotation at the effective rate $\rho_{\mathrm{eff}} = \rho(1-\kappa)$ in the mean field. Second, with $p_0 = 1$ the V0 rotation term $\rho(1-y)$ combines with exploration into $(\nu+\rho)(1-y) - \nu y$, which is umpire exploration biased toward honesty. Fresh-cohort rotation is therefore close in spirit to enforcer "education" studied by Lee et al. [9] in another model. The inheritance variants, pulses and the finite-population process differ from biased exploration.

We test six hypotheses, stated before any extension experiment was run: H1, rotation raises time-averaged honesty and damps corruption cycles; H2 (vigilance paradox), by keeping umpires honest, rotation lets prudent cooperators decline, so the remaining corruption episodes are deeper; H3, inheritance weakens the benefit of rotation; H4, rotation and a lower information cost h act as substitutes or complements; H5, conformist inheritance creates bistability and hysteresis; H6, delayed inheritance destabilizes the honest state at high κ.

3 Methods

Mean field. We integrate Eqs. (4) and (6) with LSODA (`scipy`), relative tolerance 10^{-9} and absolute tolerance 10^{-11}. Payoffs are evaluated at $x/\sum_i x_i$; this keeps the simplex invariant exactly, whereas the unnormalized form makes the simplex unstable when the mean payoff is negative. For the extension we use $\mu = \nu = 0.01$, where the base model still cycles with deep corruption and the period is short enough for long averages (Appendix C). Runs start from the initial state of Lee et al.'s Fig. 3 ($y = 1$, $x = (0.1, 0.4, 0.4, 0.1)$), discard a burn-in of 10000 time units and average over 10000 time units.

Bifurcation analysis. We derive the Jacobian of the reduced system (x_1, x_2, x_3, y) symbolically with `sympy` and check it against finite differences. Equilibria are followed in a parameter by pseudo-arclength continuation. A fold is detected by a sign change of the parameter component of the tangent and refined by solving $F = 0$, $\det J = 0$. A Hopf point is detected when the number of eigenvalues with positive real part changes by two through a complex pair, and refined by solving $F = 0$, $\mathrm{Re}\,\lambda = 0$ [17].

Delay equations. V3 is a delay differential equation. We integrate it by the method of steps: each interval of length T_{lag} is solved with `solve_ivp`, using the dense output of the previous interval for the delayed term. Local stability of an equilibrium is decided by the rightmost root of the characteristic equation $\det(\lambda I - J_0 - J_1 e^{-\lambda T_{\mathrm{lag}}}) = 0$, computed by Chebyshev collocation of the infinitesimal generator with 40 nodes and checked with 80 [18].

Individual-based model. The SI of Lee et al. states the rules that our simulation follows: an exploring player adopts one of the three other strategies at random, an exploring umpire switches type, umpires explore at rate $\nu = \mu M/N$, and under hard imitation a tie is resolved with probability 1/2 [1]. It does not state the order of updates, how role models are chosen, or how payoffs are evaluated in a finite population; the choices below are our assumptions (Appendix B). In each elementary step one individual is chosen uniformly from all $M + N$ individuals and revises its strategy as described above. A player's co-player is drawn from the other $M - 1$ players, and an umpire serves two distinct random players; payoffs are expected values given the current counts. Rotation adds replacement events at rate ρN per generation. One generation is the time in which each individual revises once on average. Because payoffs depend only on counts, the simulation tracks counts and is exact for this process. One unit

of time in Lee et al.'s Fig. 1 is one elementary step; the numbers of cycles in SI Fig. 8 agree with our model at the same settings, which supports this time unit (Appendix E). We use the two settings of Lee et al.'s Fig. 1: small ($M = 50$, $N = 10$, $\mu = 0.001$, $\nu = 0.005$, $s = 10^{10}$) with sustained cycles, and large ($M = 5000$, $N = 1000$, $\mu = 0.01$, $\nu = 0.05$, $s = 0.3$) with damped oscillations. For validation we derive the exact large-population limit of this process, $\dot{x}_i = (1-\mu)x_i \sum_j x_j \tanh[s(P_i - P_j)/2] + \mu[(1-x_i)/3 - x_i]$ and the analogous equation for y, which reduces to Eqs. (4)–(5) for weak selection after rescaling time by $s/2$ and exploration rates by $2/s$.

Rare-exploration limit. When exploration is rare and imitation is hard, the population moves between ten absorbing states. We derive the transition matrix from the payoffs: a mutant that earns more than the residents takes over (and we follow the replicator dynamics to find where the population ends), a neutral umpire mutant takes over with probability $1/N$, and a neutral player mutant takes over with probability $1/2$ if two mutants would earn more than the residents, never if they would earn less, and with probability $1/M$ if they stay neutral. These are the rules stated in the SI. The stationary distribution is computed symbolically.

Metrics and pre-registration. Before running any extension experiment we fixed the metrics, burn-in, averaging windows and decision rules for each hypothesis (file `METRICS.md`, committed first). The main metrics are the time-averaged honest fraction $\bar{y}$; the cooperator share $x_1 + x_2$; the frequency of prudent cooperators (vigilance); the cycle amplitude; and corruption episodes, defined as maximal intervals with $y < 0.5$, with their depth (peak corruption $1 - \min y$ and lowest cooperator share within the episode), duration and frequency.

Statistics. Each finite-population condition has 30 independent replicates. In the small setting each replicate runs 5000 generations of burn-in and 100000 generations of averaging; in the large setting 1000 and 5000. We report means with 95% percentile bootstrap intervals (10000 resamples). The standard error of $\bar{y}$ fell from 0.0042 with 5 replicates to 0.0016 with 30 and 0.0014 with 50, which justified 30 replicates. Time averages did not change when the window was doubled (Appendix D).

Sensitivity and software. We varied $b, c, f, B, h, A, \mu, \nu$ one at a time by $\pm 20\%$ in the mean field and s by $\pm 20\%$ in the large finite-population setting. The code is in Python (`numpy`, `scipy`, `sympy`, `numba`); it has 100 automated tests, and every number in this paper is generated from result files by a script. The full pipeline used about 17 minutes on a laptop.

4 Results

4.1 Reproduction of Lee et al. (2019)

Table 1 compares our results with the published benchmarks. With all umpires corrupt and no exploration, every trajectory converged to OD $= h/(f+B) = 0.25$ and PC $= 0.75$ (largest deviation 4.8×10^{-12} over 20 random starts); with all umpires honest, OC went to fixation. With the printed weighting of $Q(y)$, the all-corrupt case instead ends at OC $= 1.000$, which contradicts the paper's own statement.

In the small population, corruption episodes recur (Fig. 1) at a rate close to that in the published figure. In the rare-exploration limit, our derivation of the transition matrix from the payoffs agrees with the published matrix except for one entry (D4) in a row that the chain never revisits, and gives exactly the published ratio $[(7+12\lambda)N + 33\lambda + 54\lambda^2] : [7 + 48\lambda + 54\lambda^2]$, which is 2.541 for $N = 10$ and $\lambda = 1$. It also reproduces the SI's chain for cost-free information ($h = 0$) exactly. The simulated ratio at the published exploration rates depends on how the time average

Table 1: Reproduction benchmarks. Values in brackets are 95% bootstrap intervals over 30 replicates. "Read off" values were estimated by eye from the published figures.

Benchmark	Lee et al. (2019)	This work
B1: all corrupt	OD 0.25, PC 0.75	OD 0.25, PC 0.75
B2: all honest	OC $\to 1$	OC $\to 1$
B3: H:C ratio, rare exploration ($N = 10$)	$(19N + 87){:}109 = 2.54$	2.541 (independent derivation)
B3: simulated H:C ratio	about 2.15	2.030 [2.003, 2.057] (mean shares); 2.134 [2.102, 2.165] (time all honest : all corrupt)
B4: episodes per 10^5 steps (small)	about 7 (read off Fig. 1)	6.0 [6.0, 6.1]
B4: corrupt fraction (large)	0.50–0.55 (read off Fig. 1D)	0.541 [0.535, 0.545]
B4: orbit shrinkage	qualitative (Fig. 3)	Hopf at $\mu = \nu = 0.0616$ ($A = 2$), 0.0314 ($A = 0.7$)

is defined, which neither the paper nor the SI states. The ratio of the mean honest share to the mean corrupt share is 2.030 [2.003, 2.057], lower than the published 2.15. The ratio of the time with all umpires honest to the time with all umpires corrupt, the quantity that the Markov chain predicts, is 2.134 [2.102, 2.165] and agrees with it. The mean-share value is stable when the averaging window is doubled (2.032) and in 10 runs of 2.2×10^8 steps (mean 2.037), and approaches the Markov limit as exploration becomes rarer: 2.318 at $\mu = 3 \times 10^{-4}$ and 2.459 at $\mu = 10^{-4}$ (Fig. 2). Changing the burn-in, the initial state, the imitation strength, or the conventions for exploration, role models and payoffs did not bring it to 2.15 (Appendix E).

In the large population, the simulations follow the exact mean field of the process and settle at a corrupt fraction of 0.541 after damped oscillations (Fig. 3A). The mean-field equilibrium is a stable focus with eigenvalues $-0.0081 \pm 0.0089i$, which corresponds to a period of about 705 generations and a decay time of 124 generations. In the replicator-mutator equations, raising the exploration rate shrinks the limit cycle until it disappears in a Hopf bifurcation (Fig. 3B, C). We found that the time series in Lee et al.'s Fig. 3C, D match the orbit at $\mu = \nu = 0.02$ (period 136.5; peaks of OC, PC, PD and OD 0.923, 0.697, 0.438 and 0.268) rather than the orbit at 10^{-4} named in the caption, whose period is 9437 (Fig. 3D). All discrepancies are listed in Appendix B.

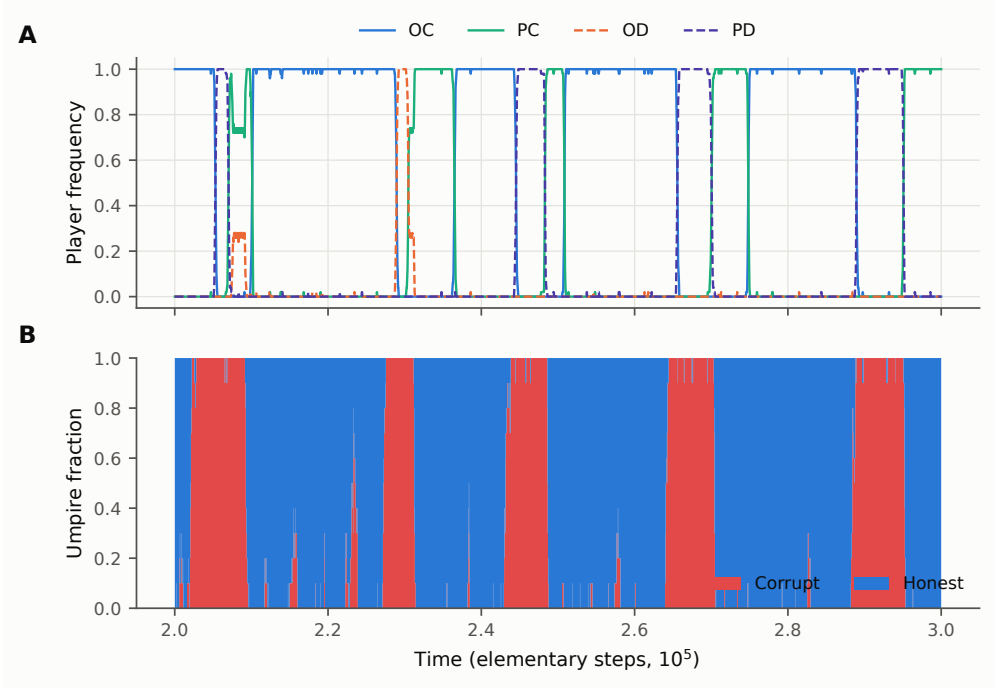


Figure 1: Sustained cycles in the small population (reproduction of Lee et al. Fig. 1A, B). (A) Frequencies of OC, PC, OD and PD. (B) Shares of honest (blue) and corrupt (red) umpires. One replicate, elementary steps 2–3 $\times 10^5$. Parameters: $b = 1$, $c = 0.5$, $f = B = 0.2$, $h = 0.1$, $A = 2$, $M = 50$, $N = 10$, $\mu = 0.001$, $\nu = 0.005$, $s = 10^{10}$.

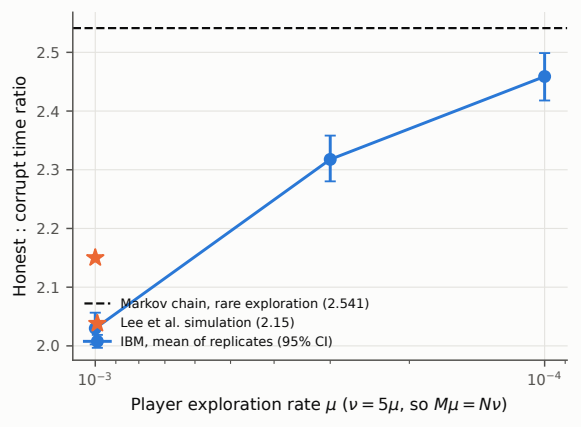


Figure 2: Honest-to-corrupt time ratio in the small population. Dots: mean over 30 replicates with 95% bootstrap intervals; each replicate runs 200000 generations at $\mu = 10^{-3}$ and proportionally longer at smaller μ, with $\nu = 5\mu$ so that $M\mu = N\nu$. Dashed line: rare-exploration Markov chain. Star: value reported by Lee et al.

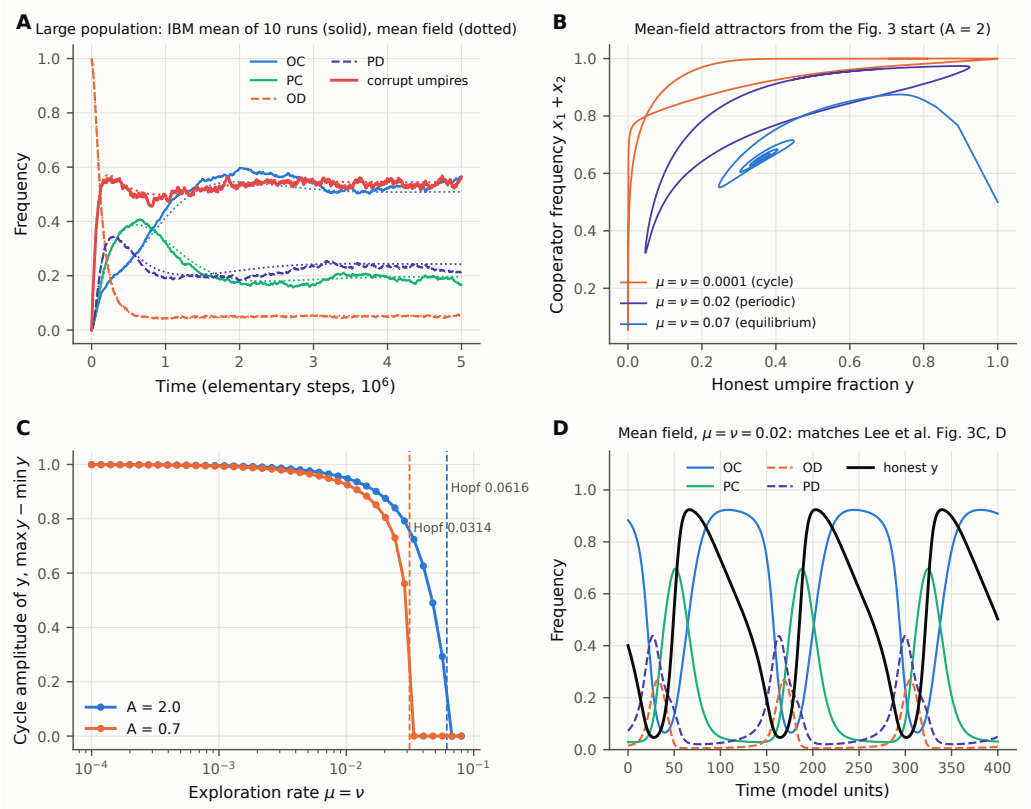


Figure 3: Damped oscillations and orbit shrinkage. (A) Large population ($M = 5000$, $N = 1000$, $\mu = 0.01$, $\nu = 0.05$, $s = 0.3$), start with all OD and all honest umpires: mean of 10 replicates (solid) and exact mean field (dotted). (B) Mean-field attractors in the plane of honest fraction and cooperator share for $\mu = \nu \in \{10^{-4}, 0.02, 0.07\}$, $A = 2$. (C) Cycle amplitude of y against $\mu = \nu$; dashed lines mark the Hopf points. (D) Mean-field time series at $\mu = \nu = 0.02$, $A = 2$, which match Lee et al.'s Fig. 3C, D.

4.2 Analytical results for rotation

Rotation removes cycles through a SNIC. Figure 4A shows the equilibria of the mean field with fresh-cohort rotation (V0, $p_0 = 1$) as a function of ρ. Without rotation, the only interior equilibrium is an unstable focus surrounded by the corruption cycle. At $\rho_c = 0.0133$ a fold creates a stable node close to the all-OC state (honest fraction 0.582, OC 0.957, PC 0.015) together with a saddle. The fold lies on the limit cycle: as $\rho \to \rho_c$ the orbit lingers near the coming equilibrium and the period grows as $(\rho_c - \rho)^\alpha$ with fitted $\alpha = -0.478$, close to the value $-1/2$ that is characteristic of a saddle-node on an invariant circle (SNIC; Fig. 4B). For $\rho > \rho_c$ all three initial states we tried went to the honest equilibrium. The unstable focus is destroyed in a second fold at $\rho = 0.126$. At the default exploration rate there is no Hopf bifurcation in ρ.

An approximation for the critical rotation rate. Near the all-OC state with small exploration, each rare strategy j is kept at a mutation-selection balance $x_j \approx \mu/(3d_j(y))$, where $d_j(y)$ is its payoff deficit against OC:

$$d_{\mathrm{PC}} = (1-y)(b-c-f)+h, \quad d_{\mathrm{OD}} = y(A-B)-(c-B), \quad d_{\mathrm{PD}} = y(b-f-B)-(c-B-h). \tag{7}$$

OD and PD can invade the OC state when y falls below $y_{\mathrm{OD}} = (c-B)/(A-B)$ or $y_{\mathrm{PD}} = (c-B-h)/(b-f-B)$; these are the two ratios compared in Lee et al.'s Fig. 3 caption. Inserting x_j into Eq. (3) gives $P_H - P_C \approx 4f(x_2 - x_4) - 2B(x_3 + x_4)$, and the umpire equation (6) has an equilibrium at y when

$$\rho = G(y) = -\frac{y(1-y)\,[P_H - P_C](y) + \nu(1-2y)}{p_0 - y}. \tag{8}$$

The honest equilibrium appears at $\rho_c \approx \min_{y > y_{\mathrm{th}}} G(y)$ with $y_{\mathrm{th}} = \max(y_{\mathrm{OD}}, y_{\mathrm{PD}})$. At the default parameters this gives 0.0137 (exact 0.0133; error 2.5%). Because G is linear in (μ, ν), the critical rate is proportional to the exploration rate: $\rho_c/\mu \approx 1.36$ for $A = 2$ and 3.35 for $A = 0.7$ when $\mu = \nu$ (Fig. 4C). For $\mu \le 0.011$ the formula is within 2.5% of continuation for $A = 2$. For $A = 2$ the two folds exist up to $\mu = \nu \approx 0.056$. Close to the base Hopf point ($\mu = \nu = 0.0616$) Hopf points also appear on the equilibrium branch (crosses in Fig. 4C), so there the route by which rotation removes the cycle changes; we did not analyze this region in detail.

Linear inheritance equals slower rotation. For V1 the whole mean-field system equals the V0 system with ρ replaced by $\rho(1-\kappa)$ (checked symbolically). Every bifurcation point of V0 at ρ_* therefore appears in V1 at $\rho_*/(1-\kappa)$, which we confirmed by continuation for both folds. The same identity holds for the pulse map, $(1-\tau)y + \tau[(1-\kappa)p_0 + \kappa y] = y + \tau(1-\kappa)(p_0 - y)$.

Dependence on the information cost. The critical rotation rate depends on h non-monotonically (Fig. 5A, B): it peaks at 0.0165 near $h = 0.04$ and falls to 0.0017 at $h = 0.28$. The reason is that y_{PD} decreases with h: when information is costly, prudent defection is less attractive, and a smaller push toward honesty keeps the OC state safe.

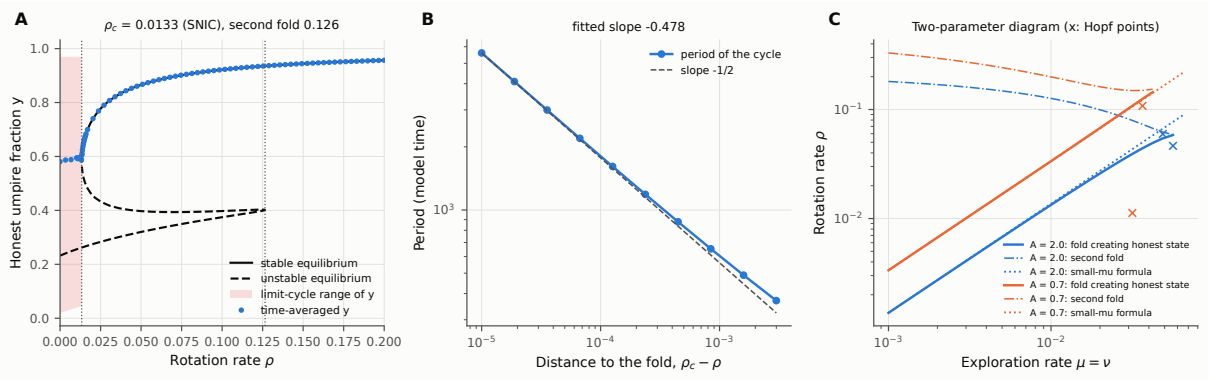


Figure 4: Mean-field bifurcations in the rotation rate (V0, $p_0 = 1$). (A) Honest fraction at the equilibria (solid stable, dashed unstable), range of y on the limit cycle (shaded) and time-averaged y (dots); dotted lines mark the folds. $\mu = \nu = 0.01$, $A = 2$. (B) Period of the cycle against the distance to the fold, log-log; dashed line slope $-1/2$. (C) Fold points (solid: the fold creating the honest state; dash-dot: the second fold), the small-exploration formula (dotted) and Hopf points (crosses) as functions of $\mu = \nu$, for $A = 2$ and $A = 0.7$.

4.3 Hypotheses

Table 2 summarizes the outcomes, which were decided by the rules in `METRICS.md`.

H1: rotation raises honesty and damps cycles (supported). In the mean field the amplitude of the cycle is unchanged until ρ_c and zero beyond it (Fig. 4A). Time-averaged honesty is nearly flat below ρ_c (0.581 at $\rho = 0$, 0.586 just below ρ_c) because the orbit spends more time near the ghost of the coming equilibrium, then rises to 0.868 at $\rho = 0.05$ and 0.957 at $\rho = 0.2$. In the small finite population, honesty rises from 0.669 [0.665, 0.673] without rotation to 0.978 [0.978, 0.979] at $\rho = 0.3$ per generation, and corruption episodes fall from 3.63 to 0.29 per 1000 generations (Fig. 6D, F). In the large population honesty rises from 0.458 to 0.876, and episodes with $y < 0.5$ disappear for $\rho \geq 0.05$. In the large population the time-averaged honesty with rotation is within 0.008 of $(\nu + \rho)/(2\nu + \rho)$, the value expected if selection between umpire types were neutral, for all $\rho \geq 0.05$.

H2: the vigilance paradox (not supported). Rotation lowers vigilance as predicted. In the small population the frequency of prudent cooperators falls from 0.210 to 0.050 at $\rho = 0.1$ and 0.006 at $\rho = 0.3$; in the mean field from 0.174 to 0.036 just below ρ_c and 0.016 at the honest equilibrium. But the remaining episodes are not deeper (Fig. 6). In the small population the lowest cooperator share during an episode rises from 0.262 [0.253, 0.271] to 0.643 [0.633, 0.653] at $\rho = 0.1$, peak corruption falls from 0.965 to 0.910, and episodes last 88.1 generations without rotation and 23.3 at $\rho = 0.3$. In the mean field, peak corruption falls from 0.980 to 0.955 and the lowest cooperator share rises from 0.245 to 0.263 as ρ approaches ρ_c. A likely reason why fewer prudent players do not lead to worse episodes is that rotation also feeds honest umpires into the episodes themselves, which shortens them before cooperation collapses fully; we did not test this mechanism directly.

H3: inheritance weakens rotation (supported). In the mean field, honesty falls with κ at every rotation rate for V1 and V3 and at most rates for V2, and the region without cycles shrinks (Fig. 7A–C). For V1 the boundary is exactly $\rho(1 - \kappa) = \rho_c$. V2 weakens rotation least, because at the honest equilibrium $y > 1/2$ and conformity then favors honesty. In the small population, honesty falls with κ for all three variants at all three rotation rates tested (Fig. 7D–F). The finite-population test of the V1 equivalence compares V1 at (ρ, κ) with V0 at $\rho(1 - \kappa)$; the difference in mean honesty was at most 0.0073, but its 95% interval excluded zero at 2 of 15

Table 2: Outcomes of the pre-registered hypotheses. MF: mean field; IBM: finite population.

Hypothesis	Outcome	Main evidence
H1: rotation raises honesty, damps cycles	supported	MF: cycles vanish at ρ_c; IBM: $\bar{y}$ 0.669→0.978, episodes 3.63→0.29 per 1000 generations
H2: vigilance paradox	not supported	PC 0.210→0.006, but episodes shallower (lowest cooperation 0.262→0.643) and shorter
H3: inheritance weakens rotation	supported	Honesty falls with κ for V1, V2, V3 in MF and IBM
H4: rotation vs. transparency	substitutes	Gain from rotation rises with h in 273 of 300 MF pairs and 29 of 32 IBM pairs
H5: conformist bistability	mixed	Bistable only for $\kappa \geq 0.95$; no loop in the pre-registered sweep
H6: delay destabilizes	not supported	Rightmost root $\leq -3.2 \times 10^{-4}$ at all 1375 points

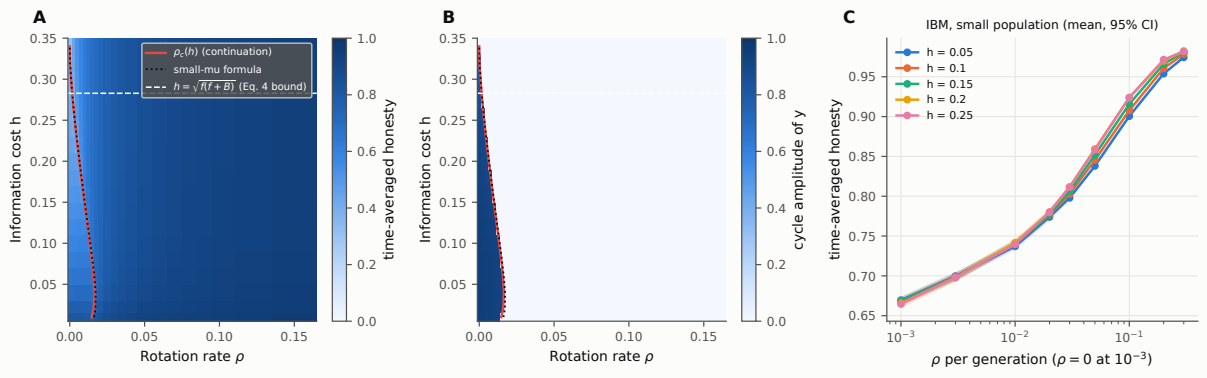


Figure 5: Rotation and the information cost h (V0, $p_0 = 1$). (A) Mean-field time-averaged honesty; red line: critical rate $\rho_c(h)$ from continuation; dotted: small-exploration formula; dashed white: the bound $h = \sqrt{f(f+B)}$ of Lee et al. (B) Mean-field cycle amplitude of y. $\mu = \nu = 0.01$, burn-in 10000, window 10000. (C) Small finite population: mean honesty with 95% bootstrap bands over 30 replicates of 100000 generations; $\rho = 0$ is plotted at 10^{-3}.

points. By the pre-registered rule the equivalence therefore does not hold exactly in the finite population; with a Bonferroni adjustment for 15 comparisons (a post-hoc check) it holds.

H4: rotation and transparency are substitutes. If rotation helps more where information is costly, the two act as substitutes. In the mean field the gain in honesty from rotation increased with h for 273 of 300 neighboring pairs in the admissible range of h (for example 0.134 at $h = 0.02$ and 0.478 at $h = 0.28$ for $\rho = 0.05$), so the pre-registered rule classifies them as substitutes for honesty. For the cooperator share the pattern is mixed (189 pairs increasing, 90 decreasing). In the small population the gain increased with h in 29 of 32 pairs, while honesty without rotation hardly depended on h (0.665 to 0.669), as the rare-exploration result of Lee et al. predicts (Fig. 5C).

H5: conformist inheritance and bistability (mixed). In the pre-registered scan over h, exploration rate, conformity exponent, κ and ρ, only 1 of 240 points showed different attractors from an honest and a corrupt start, and the pre-registered sweeps of κ showed no loop (largest gap 0.012; Fig. 8A, F). A targeted follow-up found an open bistable region, but only when newcomers almost purely copy the majority ($\kappa \geq 0.95$ at default parameters, $\kappa \geq 0.85$ with $h = 0.29$ and $k = 4$) and rotation is fast (Fig. 8B, C). A post-hoc sweep that starts from the corrupt state shows hysteresis (gap 0.95; Fig. 8D). In the finite population the time-averaged

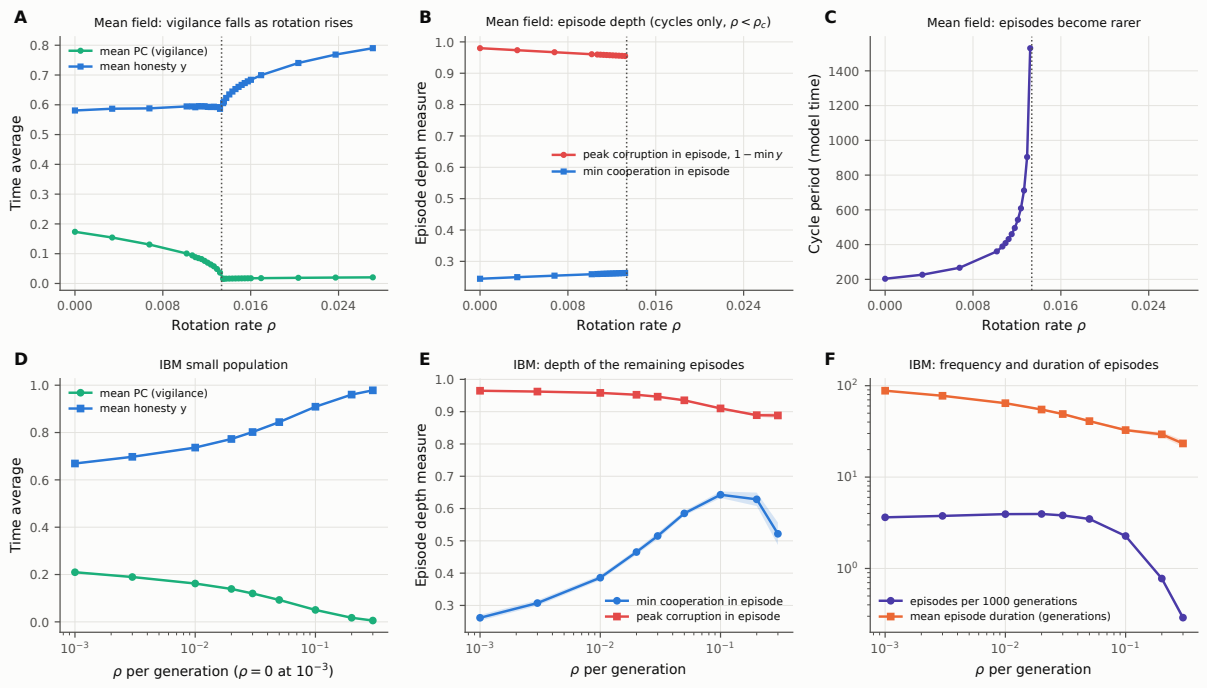


Figure 6: Vigilance and the depth of corruption episodes (V0, $p_0 = 1$). Top: mean field ($\mu = \nu = 0.01$, $A = 2$); dotted line marks ρ_c; depth and period are shown only where a cycle exists. Bottom: small finite population, means with 95% bootstrap bands over 30 replicates of 100000 generations. An episode is a maximal interval with $y < 0.5$; in the finite population gaps shorter than 5 generations are merged and episodes shorter than 5 generations dropped.

honesty did not depend on the starting state at these points, for $N = 10$ and $N = 100$ (Fig. 8E), so noise switches between the two states. The hypothesis is therefore only partly supported.

H6: delayed inheritance (not supported). The equilibria of V3 are those of V1, so the honest equilibrium exists when $\rho(1 - \kappa) > \rho_c$. Delay could in principle destabilize it. Across 1375 combinations of $\rho \in \{0.05, 0.1, 0.2\}$, κ and T_{lag} from 0 to 400, the rightmost characteristic root had real part at most -3.2×10^{-4}, and a check with twice the collocation nodes gave the same picture (Fig. 9). Longer delays only slowed the return to the honest state. Direct simulations of the delay equation agreed.

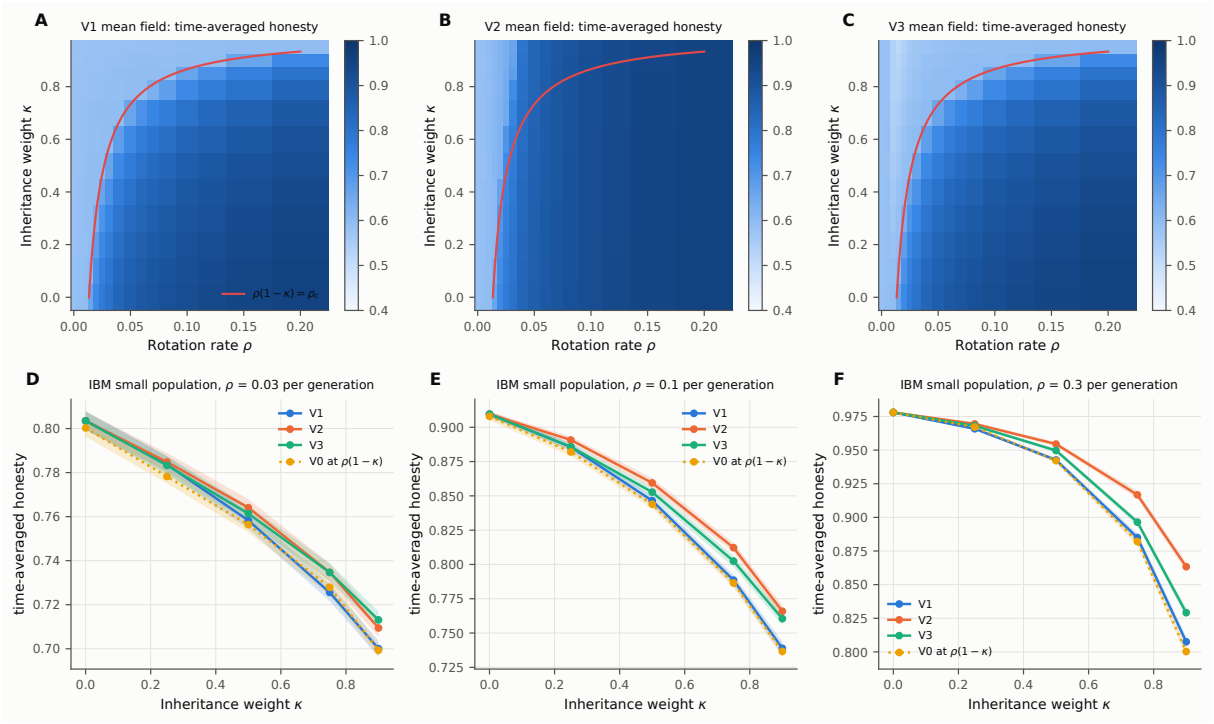


Figure 7: Inherited integrity. (A–C) Mean-field time-averaged honesty for V1, V2 ($k = 2$) and V3 ($T_{\text{lag}} = 1/\rho$) over rotation rate and inheritance weight; red line: $\rho(1-\kappa) = \rho_c$. $\mu = \nu = 0.01$. (D–F) Small finite population at $\rho = 0.03$, 0.1 and 0.3 per generation: mean honesty with 95% bootstrap bands over 30 replicates; dotted: fresh rotation at $\rho(1-\kappa)$.

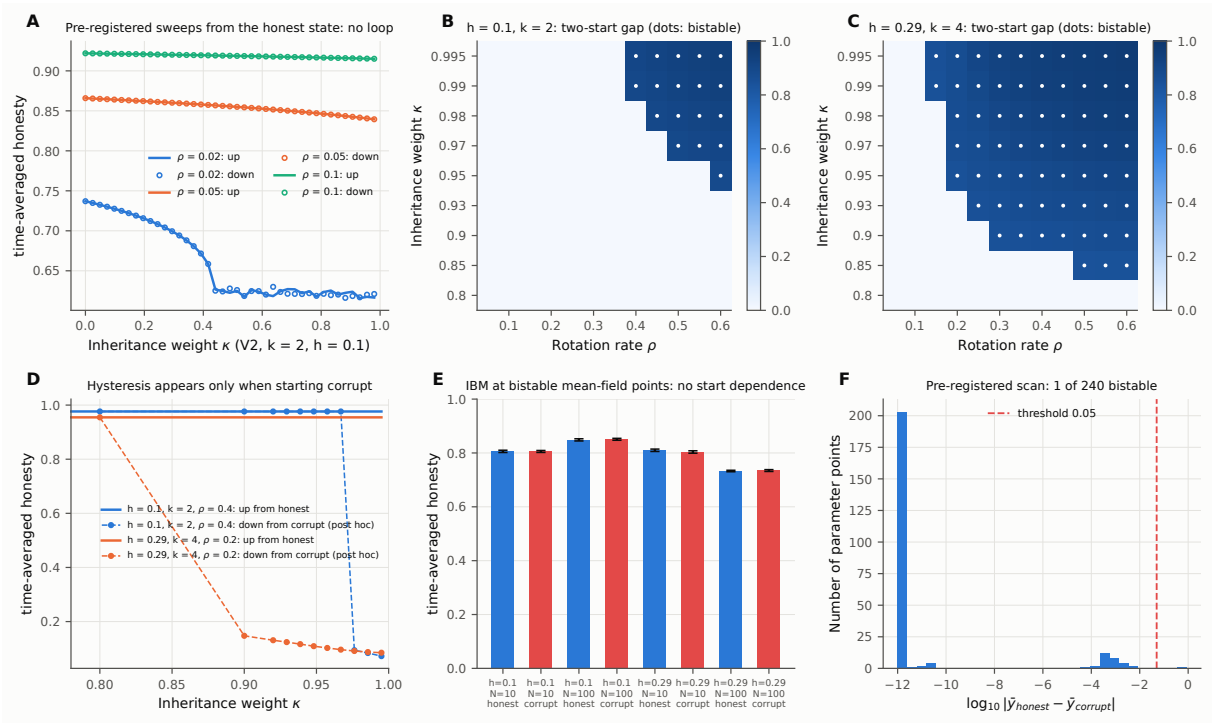


Figure 8: Conformist inheritance (V2). (A) Pre-registered sweeps of κ up and down, continuing the state, at three ρ ($h = 0.1$, $k = 2$, $\mu = \nu = 0.01$). (B, C) Difference in time-averaged honesty between honest and corrupt starts over (ρ, κ); dots mark points where it exceeds 0.05. (D) Sweeps up from the honest state and down from the corrupt state (post hoc). (E) Small and medium finite populations at bistable mean-field points, honest and corrupt starts, means with 95% intervals over 30 replicates. (F) Distribution of the two-start difference in the pre-registered scan (log scale).

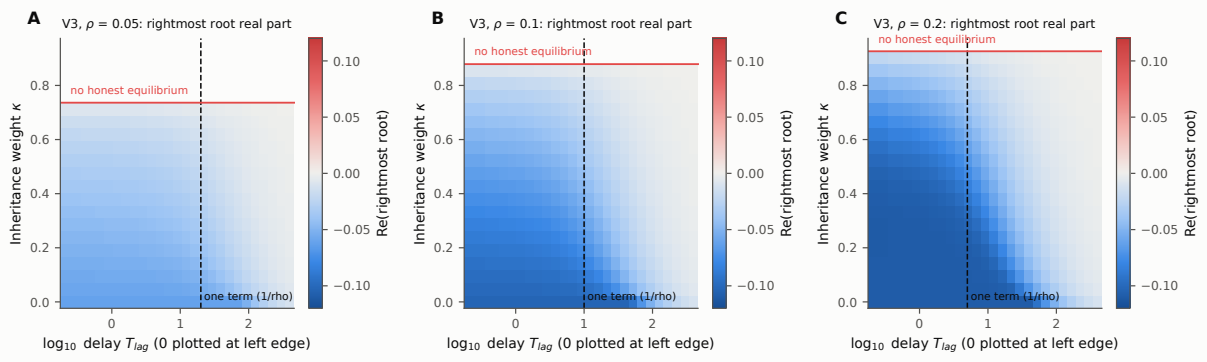


Figure 9: Delayed inheritance (V3). Real part of the rightmost characteristic root of the honest equilibrium over delay and inheritance weight, for $\rho = 0.05$, 0.1 and 0.2 ($\mu = \nu = 0.01$). Blue: stable. Dashed line: one term, $T_{\text{lag}} = 1/\rho$. Above the red line no honest equilibrium exists ($\rho(1-\kappa) < \rho_c$).

4.4 Robustness

Replacing whole cohorts at once is more effective than gradual replacement at the same average rate. In the mean field at $\rho = 0.02$, continuous rotation gives honesty 0.737 and replacing all umpires every $1/\rho$ time units gives 0.825. In the small population at $\rho = 0.03$ the values are 0.802 [0.798, 0.806] and 0.899 [0.897, 0.900] (Appendix Fig. S3). When newcomers are not reliably honest, rotation needs to be faster and can backfire: ρ_c rises to 0.0174, 0.0302 and 0.0739 for $p_0 = 0.9$, 0.75 and 0.6, and with $p_0 = 0.6$ moderate rotation lowers honesty to 0.484. With a smaller fine ($A = 0.7$) the equilibrium curve has the same two folds and no Hopf point, with $\rho_c = 0.0336$. In one-at-a-time variations of $\pm 20\%$ that respect the parameter conditions, ρ_c ranged from 0.0058 to 0.0254 and honesty at $\rho = 0.05$ from 0.845 to 0.889 (Appendix Fig. S2). Rotation raised honesty at all population sizes tested (Fig. 10B); larger populations had lower honesty without rotation (0.673 for $M = 50$, 0.616 for $M = 400$) and needed faster rotation for the same effect.

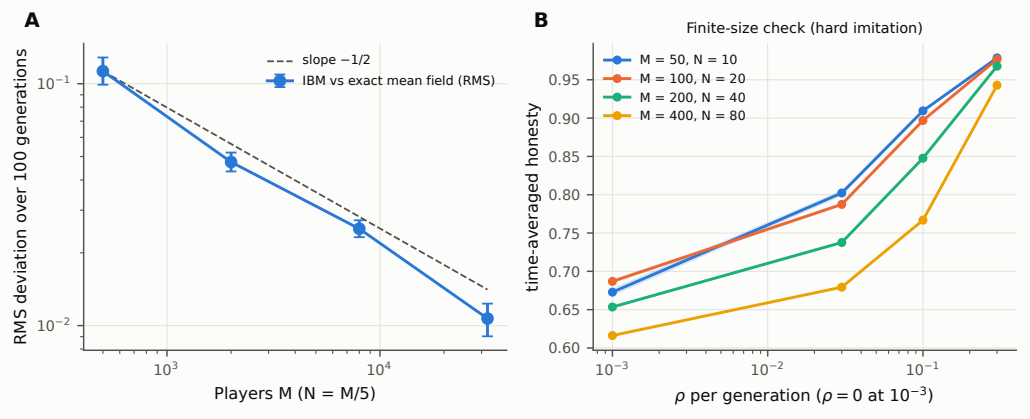


Figure 10: Validation and finite size. (A) Root-mean-square deviation between the finite population and its exact mean field over 100 generations, against M ($N = M/5$, large-population rates, 8 replicates, 95% intervals); dashed: slope $-1/2$. (B) Mean honesty against rotation rate for $M \in \{50, 100, 200, 400\}$, $N = M/5$, small-population rates, 30 replicates, 95% bands.

5 Discussion

Lee et al. showed that institutional corruption need not settle at a fixed level: when watching the watchers is costly, honesty and corruption chase each other in cycles [1]. We asked what term limits do to this cycle. In the model, regular replacement of umpires by honest newcomers removes the cycle once the replacement rate passes a critical value. Below that value the cycle survives but slows down, and the average level of honesty barely changes. Above it the institution settles in a state where nearly all players cooperate optimistically and few check their umpires. The transition is a saddle-node on the cycle rather than a Hopf bifurcation, which has a practical reading: the system does not move smoothly from small oscillations to stability. Instead, corruption episodes become rarer and rarer as the rotation rate rises and then stop. The critical rate is proportional to how often umpires explore corrupt behavior, so the model says that the turnover needed to keep an institution honest scales with the rate at which its members are tempted to try corruption.

The vigilance paradox we expected did not appear. Rotation did make players less vigilant, but the episodes of corruption that remained were milder and shorter, because the same newcomers that keep the institution honest also cut episodes short. Within this model, then, the loss of vigilance is not dangerous as long as the supply of honest newcomers continues. This conclusion depends on that supply. When newcomers inherit the integrity of the body they join, the benefit of rotation shrinks, and linear inheritance is exactly equivalent to slower rotation. When newcomers are not reliably honest, rotation can even lower honesty. Conformist inheritance can lock an institution into corruption only in an extreme regime where newcomers almost purely follow the majority, and this lock-in did not survive noise in our finite-population simulations.

These results suggest implications for real institutions, stated here as model implications and not as empirical claims. First, the value of term limits depends on what newcomers bring: training or selection that makes newcomers reliably honest matters as much as the rotation rate. Second, mentoring by incumbents transmits whatever the incumbents are, so in a body with a corrupt majority it works against rotation. Third, replacing whole cohorts at once, as in annual elections of an entire board, was more effective in the model than staggered replacement at the same average rate. Student organizations, where leadership often changes every year, are a natural setting in which these predictions could be examined; the author's experience with the regulations governing such organizations at ITB motivated this question, but the model has not been tested against data from them.

Our work relates to several strands of research. Enforcer "education" that raises the trial

rate of honest conduct curbs corruption in a model of illegal logging [9], and higher exploration rates of enforcers reduce corruption in a model with optional external supervision [11]; in the mean field, fresh-cohort rotation is formally the same mechanism, and our contribution is the bifurcation structure, the inheritance variants and the finite-population analysis. Staff rotation has been tested as an anti-corruption policy in laboratory experiments [13, 14], and empirical studies relate corruption to term limits and re-election incentives [15, 16]. Our model omits elections and reputation, so it does not speak to those incentive effects.

The study has limitations. The SI of Lee et al. does not state the order of updates or how the reported ratio was averaged, so these parts of our individual-based model are assumptions. They reproduce the published figures and the cycle counts of SI Fig. 8, and the published ratio 2.15 is matched under one definition of the time average (2.134) but not under the ratio of mean shares (2.030). The rotation rate is measured in model time in the mean field and in generations in the simulations, and these units are not directly comparable because hard imitation is not the weak-selection limit. Populations are well mixed; players and umpires learn at the same per-capita rate; the inheritance rules are our own simple choices; and the game involves two players. The mean-field default exploration rate (0.01) is higher than the orange orbit of Lee et al. (10^{-4}); we checked 10^{-3} and 0.02 and found the same picture with ρ_c proportional to μ.

Several extensions would be natural: a formal asymptotic derivation of ρ_c; heterogeneous term lengths; mentoring on networks rather than in a well-mixed body; elections in which players choose umpires; and an analysis of pulsed rotation as a periodically forced system, which could explain why whole-cohort replacement works better.

Data and code availability

All code, configuration files, results (with configuration hashes and git commits) and the scripts that generate every figure and number are available at [repository URL placeholder].

Acknowledgements

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A Derivations

Umpire income. With x the strategy frequencies and pairs drawn independently, an honest umpire earns $2f$ unless a PD is present, so $P_H = 2f(1-x_4)^2$. A corrupt umpire earns nothing if a PC is present; otherwise it earns $2f$ plus B per defector, so $P_C = 2f(1-x_2)^2+2B(x_3+x_4)(1-x_2)$. Using $2 - x_2 - x_4 = 1 + x_1 + x_3$ on the simplex gives Eq. (3). We verified this by enumerating all ordered strategy pairs with a rule-based implementation of a single game.

Linear inheritance. $\rho[(1-\kappa)p_0 + \kappa y - y] = \rho(1-\kappa)(p_0 - y)$, so the V1 system is the V0 system with $\rho \to \rho(1-\kappa)$; for pulses $(1-\tau)y + \tau[(1-\kappa)p_0 + \kappa y] = y + \tau(1-\kappa)(p_0 - y)$.

Invasion thresholds. At the OC vertex the payoff of OC is $b - c - f$, that of OD is $y(b - f - A) + (1-y)(b - f - B)$ and that of PD is $-yh + (1-y)(b - f - B - h)$. OD does better than OC when $y < y_{\mathrm{OD}} = (c - B)/(A - B)$ and PD when $y < y_{\mathrm{PD}} = (c - B - h)/(b - f - B)$. At base parameters $y_{\mathrm{PD}} = 0.333$; $y_{\mathrm{OD}} = 0.167$ for $A = 2$ and 0.600 for $A = 0.7$. As honesty drifts down along the OC edge, the strategy with the larger threshold invades first: PD for $A = 2$ and OD for $A = 0.7$. This explains which corner the cycle visits.

Critical rotation rate. For a rare strategy j near the OC vertex, $\dot{x}_j \approx -d_j(y)x_j + \mu/3$, so $x_j \approx \mu/(3d_j)$. To first order in x_j, $P_H - P_C = 4f(x_2 - x_4) - 2B(x_3 + x_4)$, which gives Eq. (8). The honest equilibrium exists for $\rho \geq \min_{y > y_{\mathrm{th}}} G(y)$. The approximation requires $d_j \gg \mu$, which fails close to y_{th}, but the minimum lies at $y \approx 0.583$, well above the thresholds.

Neutral umpire selection. If $P_H = P_C$, Eq. (6) with $p_0 = 1$ gives $\nu(1-2y) + \rho(1-y) = 0$, so $y = (\nu + \rho)/(2\nu + \rho)$.

B Reproduction notes and resolved typos

ID	Location	Issue	Resolution
D1	Methods	$Q(y) = y_1 Q_C + (1-y_1)Q_H$ with y_1 the honest fraction	$Q(y) = y_1 Q_H + (1-y_1)Q_C$; the printed form fails B1
D2	Methods	Honest mixed state called "OD+PD (a mixture of OD and PC)"	It mixes OD and PD (Fig. 2 caption, Q_H)
D3	Fig. 3 caption	"8 or 9, respectively" reversed relative to the figure	With $A = 2$ the orbit passes state 9, with $A = 0.7$ state 8 (Appendix A)
D4	Transition matrix	Row 3 omits the transition to state 8, although the SI text says corrupt umpires invade state 3	State 3 is transient; no effect
D5	(withdrawn)	We had flagged the $\mu/6$ entry in row 4	The SI explains it by the two-mutant rule; our derivation now uses it
D6	Fig. 3C, D	Caption names the $\mu = \nu = 10^{-4}$ orbit	Panels match $\mu = \nu = 0.02$
D7	SI Fig. 9	Axis labels swapped relative to the caption	The caption is right (Appendix E)
U1	SI	Individual-based update order, role models, payoffs not stated	Assumed: random sequential updates, role model from own population, expected payoffs; time unit supported by SI Fig. 8

The time unit was inferred from Lee et al.'s Fig. 1: with one individual updating per elementary step, entry rates of new strategies are equal in both populations when $M\mu = N\nu$, as the caption states, and the number of episodes per 10^5 steps matches the figure. SI Fig. 8 supports the same time unit quantitatively (Appendix E). All signs of the payoff matrices were read from the PDF at 300 dpi; the PMC web version lost minus signs.

Symbol	Value	Source
b, c, f, B, h, A	1, 0.5, 0.2, 0.2, 0.1, 2 (or $A = 0.7$)	Lee et al., Fig. 1, Fig. 3
Small setting	$M = 50$, $N = 10$, $\mu = 0.001$, $\nu = 0.005$, $s = 10^{10}$	Lee et al., Fig. 1A, B
Large setting	$M = 5000$, $N = 1000$, $\mu = 0.01$, $\nu = 0.05$, $s = 0.3$	Lee et al., Fig. 1C, D
Mean-field exploration	$\mu = \nu = 0.01$ (also 10^{-3}, 0.02)	this work
Rotation	$\rho \in [0, 0.2]$ (mean field), $[0, 0.3]$ per generation (finite)	this work
Inheritance	$\kappa \in [0, 0.995]$, $k = 2$ (also 4), $p_0 = 1$ (also 0.9, 0.75, 0.6)	this work
Delay	$T_{\text{lag}} = 1/\rho$ (default), 0 to 400	this work

C Parameters

The base mean-field cycle has period about 9437 at $\mu = \nu = 10^{-4}$, 203.5 at $\mu = \nu = 0.01$ and 136.5 at 0.02; the Hopf point is at 0.0616.

D Numerical and statistical checks

Tightening the tolerances tenfold and switching to an explicit Runge-Kutta method changed honesty, amplitude and minimum cooperation by at most 1.1×10^{-8} at 16 test points. On a grid eight times finer in a sub-region (ρ step 0.00125), the first rotation rate with an equilibrium was within 0.0012 of the continuation value of ρ_c. The finite-population simulation agrees with its exact mean field with an error that decreases as $M^{-1/2}$ (0.113 at $M = 500$, 0.011 at $M = 32000$). With rotation rate set to zero, the code reproduces the base model bit for bit. Doubling the averaging window left the honest-to-corrupt ratio unchanged (2.032). The test suite checks payoffs against a rule-based game, the symbolic Jacobian against finite differences, the delay solver against an exact solution, and the V1 equivalence symbolically.

E Checks against the supplementary information

We obtained the SI of Lee et al. from an archived copy of the publisher's supplementary file and compared every rule it states with our model (file `SI_EXTRACT.md` in the repository). The SI does not describe a different individual-based procedure, so the results in this paper did not change; the derivation of the transition matrix was corrected to the SI's rules for neutral mutants.

Honest-to-corrupt ratio. Fig. S4A tests one possible cause of the difference between our mean-share ratio and the published 2.15 at a time. A burn-in of 50000 generations gave 2.003 [1.976, 2.031], starting from all OC gave 2.030, and $s = 1000$ gave 2.029. Weaker imitation ($s = 10$) gave 2.422, exploration that may keep the current strategy 2.512, role models that may be oneself 2.010, payoffs including self-play 2.018, and one player and one umpire updating per step 1.266. None of these is consistent with both the SI's rules and the published value. Run length does not explain it either: 10 runs of 2.2×10^8 steps gave 2.037 (SD 0.025, maximum 2.075), while 300 windows of 10^5 steps gave mean 2.39 (SD 1.13), so a single short window is too noisy to decide. The time with all umpires honest over the time with all umpires corrupt gave 2.134 [2.102, 2.165], and the time with an honest majority over the time with a corrupt majority gave 2.110 [2.080, 2.139].

Time unit. SI Fig. 8 shows single simulation runs of 5×10^6 time units for $M = 3000$, 2000, 1000 ($N = M/5$) at $s = 10$ and $s = 1$. We counted the cycles in the published panels by script and simulated the same settings (four replicates). For $s = 10$ the counts were 17, 25, 53 in the SI and 18.0, 25.8, 51.2 in our model (at most 6% apart). For $s = 1$ they were 6, 6, 14 and 6.0, 8.2, 17.2 (ours 0% to 38% higher) (Fig. S4B). A time unit of one generation instead of one elementary step would change these counts by a factor of $M + N$.

Mean-field periods. At 8 label positions in SI Fig. 9B our mean-field periods differ from the labels by 2.5% to 5.6%, except at the label 400 (20%), where the contours crowd (Fig. S4C). The SI's figure has its axis labels swapped relative to its caption (D7); the caption's assignment fits our periods.

Cost-free information. For $h = 0$ the SI gives a closed-form ratio, 27.76 for $N = 10$, $M = 50$. Our derived chain reproduces it exactly. Our simulation gives 7.90 [7.76, 8.04] at $\mu = 10^{-3}$ and 15.32 [14.80, 15.85] at $\mu = 3 \times 10^{-4}$, rising toward the limit as exploration becomes rarer.

F Literature search

We searched the web (October 2026) for evolutionary-game models of corruption that include rotation, term limits or turnover of enforcers. We found none. The closest work studies

enforcer education [9], enforcer exploration [11], corrupt enforcers in pool punishment [8] and supervision of umpires [10]. Rotation itself has been studied experimentally [13, 14] and term limits empirically [15, 16].

G Supplementary figures

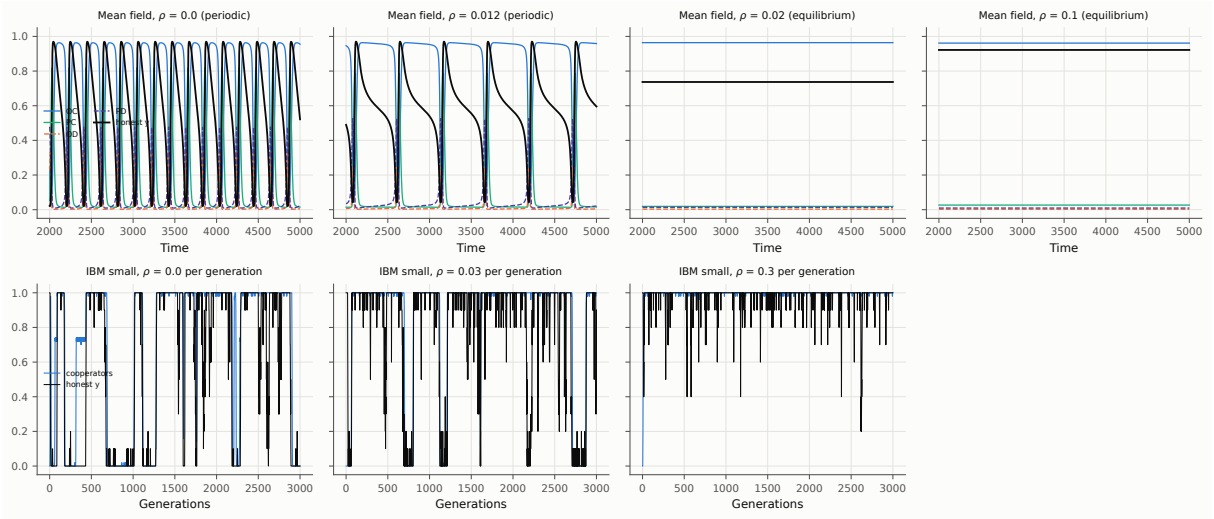


Figure S1: Time series. Top: mean field at $\rho = 0$, 0.012, 0.02 and 0.1 ($\mu = \nu = 0.01$). Bottom: one small-population replicate at $\rho = 0$, 0.03 and 0.3 per generation.

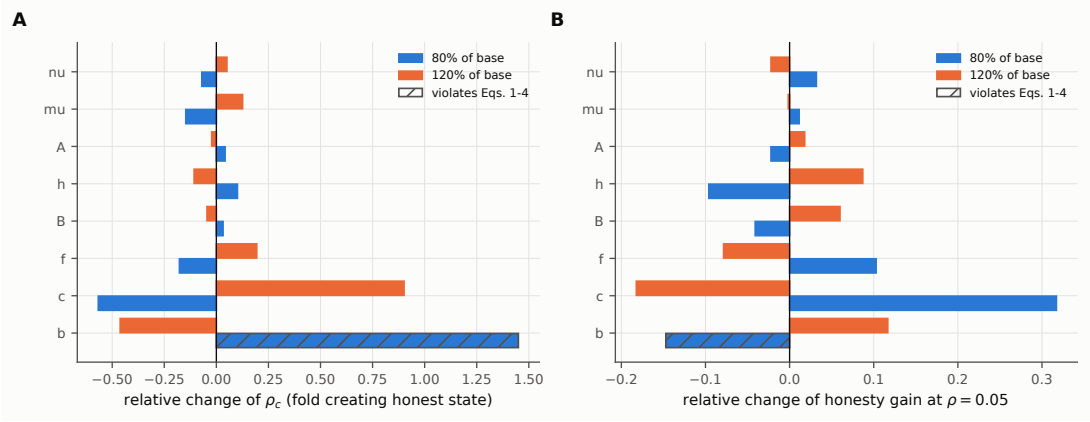


Figure S2: One-at-a-time sensitivity (±20%) of (A) the critical rotation rate and (B) the gain in honesty at $\rho = 0.05$, mean field. Hatched bars violate the parameter conditions of Lee et al.

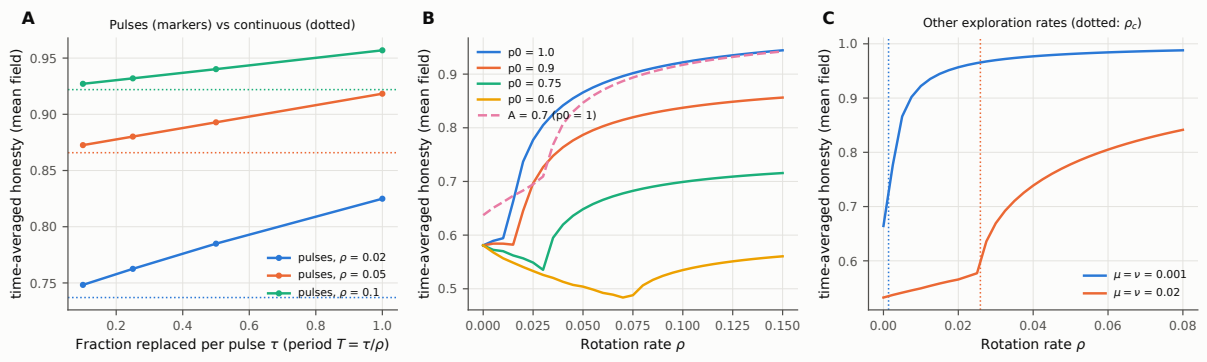


Figure S3: Robustness. (A) Pulsed rotation (markers) against continuous rotation (dotted) at the same average rate, mean field. (B) Newcomer honesty $p_0 < 1$ and $A = 0.7$. (C) Exploration rates $\mu = \nu = 10^{-3}$ and 0.02; dotted lines mark ρ_c.

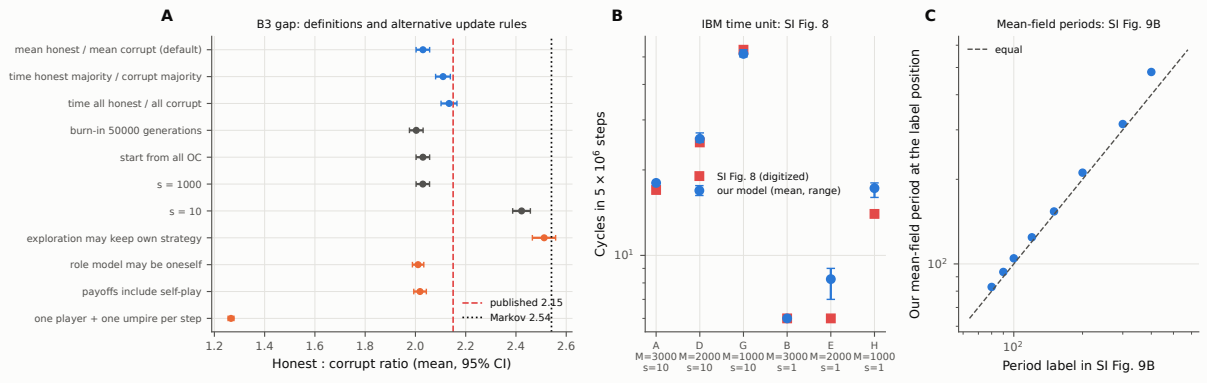


Figure S4: Checks against the SI of Lee et al. (A) Honest-to-corrupt ratio in the small population (30 replicates, mean and 95% bootstrap interval) under three definitions of the time average (blue) and with one change to the default model each (gray: run settings; orange: alternative update conventions). Dashed: published value; dotted: Markov limit. (B) Cycles in 5×10^6 elementary steps: digitized from SI Fig. 8 (squares) and our model at the same settings (circles: mean and range of four replicates). (C) Our mean-field period at the positions of the period labels in SI Fig. 9B, against the label value.